Discrete-Event Simulation of Critical Care Unit Capacity

Abstract

This study develops a discrete-event simulation (DES) model to evaluate patient flow and capacity constraints in a 24-bed Critical Care Unit (CCU), based on the framework of Griffiths et al. (2010). The objective is to assess system performance under current capacity and explore the impact of capacity expansion on congestion and access to care.

The model is implemented in Python using SimPy and incorporates six patient arrival streams. Unplanned arrivals are modelled using exponential inter-arrival times, while elective admissions follow a grouped arrival process reflecting scheduled surgical sessions. Length of stay is represented using lognormal distributions to capture variability and heavy-tailed behaviour. The system is modelled as a capacity-constrained queue, with unplanned patients waiting and elective patients cancelled when no bed is available. Simulation experiments were conducted over a one-year horizon with 472 replications.

Results indicate high baseline utilisation (87%), prolonged waiting times (45.02 hours), and substantial elective cancellations (421 annually). Increasing bed capacity significantly reduces congestion, with the largest improvements observed within the first 5–10 additional beds, followed by diminishing returns.

These findings highlight the sensitivity of critical care systems to capacity constraints and the importance of adequate resource planning.

Introduction

Critical care units (CCUs) are among the most resource-intensive components of the National Health Service (NHS), providing care for critically ill patients with complex needs. Demand for CCU beds in the UK is projected to grow by 4–5% annually, placing increasing pressure on systems that already operate near capacity [5]. A key consequence is the cancellation of elective surgeries when post-operative CCU beds are unavailable. In the first quarter of 2018 alone, 25,475 operations were cancelled on the day of surgery in England, highlighting the scale of the problem [4].

Managing CCU capacity is challenging due to competing demands from unplanned emergency admissions and elective patients. While queueing theory has been used to support capacity planning, its practical application remains limited, and reliance on fixed occupancy targets can lead to inefficient resource allocation [5]. Discrete-event simulation (DES) provides a flexible alternative by modelling system-specific variability and enabling scenario analysis.

This study replicates the DES model of Griffiths et al. (2010) [1], which simulates patient flow across six arrival streams and incorporates stochastic variability in arrivals and length of stay. The research examines how changes in bed capacity affect key performance measures, including waiting times, cancellations, and utilisation (figure 1). Due to incomplete reporting in the original study, assumptions were required, including reconstructing elective arrivals using digitised data and modelling length of stay using lognormal distributions.

Methodology

This study develops a discrete-event simulation (DES) model of patient flow in a 24-bed Critical Care Unit (CCU), based on the framework proposed by Griffiths et al. (2010). DES was selected as it allows for modelling the stochastic and time-dependent nature of healthcare systems, particularly variability in patient arrivals and length of stay (LOS), and enables the evaluation of alternative capacity scenarios.

The model was implemented in Python using the SimPy library. Patient arrivals are represented through six streams: Accident and Emergency (A&E), ward transfers, emergency surgery, transfers from other hospitals, X-ray referrals, and elective surgery. Unplanned arrivals are modelled using exponential inter-arrival time distributions, reflecting random demand, while elective arrivals were reconstructed by digitising published inter-arrival time charts using WebPlotDigitizer and sampled from a grouped continuous empirical distribution to capture the observed clustering associated with scheduled surgical sessions. Length of stay is modelled using lognormal distributions derived from empirical data, allowing the model to capture the right-skewed and heavy-tailed nature of LOS in critical care. The system operates under a capacity constraint: patients are admitted if a bed is available; otherwise, unplanned patients join a queue and wait until a bed becomes available, while elective patients are cancelled at arrival. Following discharge, each bed undergoes a fixed cleaning period of five hours before becoming available.

To ensure reliable performance estimates, a warm-up period of 500 time units was applied to minimise initialisation bias. The model was then simulated over a one-year horizon (8760 hours). Each scenario was replicated 472 times using independent random seeds to ensure statistical stability. Results are reported as mean values across replications. A capacity expansion analysis was conducted by increasing the number of beds beyond the baseline of 24 in incremental steps. System performance was evaluated using key indicators, including waiting time, queue size, bed utilisation, and planned cancellations. As a 5% precision level could not be achieved for all performance metrics, the number of replications was increased to 500. However, a precision level of approximately 8% was attained, and the highest stable number of replications (472) was selected for the final analysis.

Results

The simulation results indicate that the Critical Care Unit (CCU) operates under substantial pressure in the baseline configuration of 24 beds. A warm-up period of 500 time units was applied to minimise the influence of initial conditions, after which the model was run across multiple replications to estimate average system performance.

Under baseline conditions, the unit exhibited high utilisation, with a mean bed utilisation of 0.87. This was accompanied by a mean of 421.44 planned cancellations annually, a mean waiting time of 45.02 hours for unplanned patients, and an average queue size of 5.63 patients. The system also recorded an average of 2219.70 admissions and 2198.08 discharges, indicating sustained demand that exceeds available capacity.

Substantial variability was observed across replications. Waiting times ranged widely, and planned cancellations varied considerably between runs. This variation reflects the stochastic nature of patient arrivals and length of stay. It indicates that system performance is highly sensitive to fluctuations in demand under high-utilisation conditions. Despite this variability, the overall pattern remains consistent: the CCU experiences persistent queues, long delays, and frequent cancellations at baseline capacity.

Capacity Expansion

To assess the impact of increasing bed capacity, four additional scenarios were evaluated by incrementally increasing the number of beds.

Table 1: Performance across capacity scenarios

Metric	24 Beds	+5 Beds	+10 beds	+15 beds	+20 Beds
Waiting time (hours)	45.02	3.56	0.32	0.03	0.00
Queue size	6.17	0.41	0.04	0.00	0.00
Bed utilization (%)	87	76	66	58	51
Planned cancellations	383.35	121.25	21.80	2.37	0.18
Total admissions	2037.39	2309.16	2409.45	2429.10	2431.29

Impact of Capacity on System Performance

The scenario analysis demonstrates a strong improvement in all key performance indicators as bed capacity increases. In the baseline scenario, the system experiences prolonged waiting times, large queue sizes, and high levels of planned cancellations, indicating that demand exceeds available resources.

Adding five beds produces a substantial improvement. Mean waiting time decreases sharply, queue size falls to near-zero levels, and planned cancellations are reduced by more than two-thirds. These findings suggest that moderate increases in capacity can significantly alleviate congestion and improve access to care.

Further increases in bed capacity yield additional improvements, although with diminishing returns. With 10 additional beds, waiting times become negligible and cancellations are reduced to minimal levels, indicating that the system is no longer capacity-constrained. Increasing capacity beyond this point results in only marginal gains, as most unmet demand has already been absorbed. At the same time, bed utilisation declines steadily as capacity increases, reflecting the transition from a highly constrained system to one with increasing operational slack. This highlights a trade-off between efficiency and accessibility, where reducing delays and cancellations requires lower average utilisation.

These patterns are further illustrated in **Figure 2**, which shows the distribution of performance metrics across scenarios. Under the baseline configuration, waiting times and queue sizes exhibit high variability with several extreme values, indicating an unstable and highly congested system. As capacity increases, both the magnitude and variability of these metrics decrease rapidly, with waiting times and queues approaching zero beyond 10 additional beds. In contrast, bed utilisation declines steadily across scenarios. This visual evidence reinforces the nonlinear relationship between capacity and performance, with the greatest improvements observed between the baseline and the addition of 5–10 beds, followed by diminishing returns.

Discussion

This study developed a discrete-event simulation model of a Critical Care Unit (CCU) based on the framework proposed by Griffiths et al. [1]. The model produced a baseline occupancy of 87% and approximately 390 elective cancellations annually under a 24-bed configuration, compared to 82% occupancy and 57 cancellations reported in the original study [1]. While the overall conclusion that 24 beds are insufficient to meet demand is consistent, the divergence in quantitative outputs highlights key challenges in reproducing published simulation models.

A primary limitation arises from the under-specification of modelling assumptions in the original study. Although Griffiths et al. describe elective arrivals as time-dependent and provide an approximate normal distribution for arrival times [1], the operational implementation of this process is not fully specified. In particular, session sizes, scheduling structure, and the mechanism used to generate clustered arrivals are not clearly defined. This is critical, as arrival processes are a dominant driver of system dynamics in discrete-event simulation, particularly in healthcare systems with time-dependent demand [2]. Elective scheduling strongly shapes arrival clustering, which in turn drives congestion and cancellation rates in capacity-constrained systems. Differences in how this process is implemented are therefore likely to contribute to the higher cancellation rates observed in this model.

A second source of discrepancy relates to distributional assumptions. The original study utilised hyperexponential distributions for several length-of-stay (LOS) groups, but did not report the full parameterisation required for replication, such as phase probabilities and rate parameters [1]. In the present model, LOS was represented using lognormal distributions as a simplification of this

multi-phase structure. While this improves transparency and reproducibility, it reduces flexibility in capturing the high variability and extreme values characteristic of LOS data. LOS distributions are known to exhibit heavy-tailed behaviour and high variability, which are key determinants of congestion in healthcare systems [4]. Consequently, variation in LOS modelling is likely to contribute directly to the higher occupancy and cancellation rates observed.

Further divergence arises from incomplete specification of queueing rules and operational logic. Although emergency patients are prioritised over elective cases [1], the queueing discipline is only partially specified. The original study does not define ordering within queues, handling of simultaneous arrivals, or detailed bed allocation logic. Additionally, implementation-level assumptions such as discharge timing and cleaning processes require interpretation, as these are not fully parameterised. In discrete-event simulation, such implementation details can significantly affect system behaviour, particularly in high-utilisation environments where small differences in event timing propagate through the system [5].

Methodological simplifications in the present model also contribute to observed differences. The model enforces a strict 24-bed capacity, whereas the original study allows for occasional use of additional unfunded beds during peak demand, with up to 29 beds occupied in practice [1]. This effectively increases realised capacity and provides a buffer against demand surges. Excluding this flexibility results in a more constrained system and is likely to contribute to the higher cancellation rates observed. This simplification may not fully capture time-dependent discharge patterns associated with clinical practices such as ward rounds, further increasing congestion within the system.

These findings highlight the sensitivity of healthcare systems operating near capacity thresholds. The discrepancy between this model and the original study therefore reflects both structural uncertainty and inherent system sensitivity, rather than model error. More broadly, this study underscores the challenges of reproducibility in healthcare simulation research. Incomplete reporting of model assumptions limits the ability to replicate results reliably [6] The differences observed here demonstrate that approximate behavioural alignment, supported by logical system dynamics and sensitivity analysis, provides stronger evidence of validity than exact numerical replication.

Conclusion

Despite quantitative differences, the model successfully reproduces the key finding that a 24-bed CCU operates under significant capacity pressure. The divergence in results is primarily attributable to under-specification of elective scheduling, differences in distributional assumptions, and the exclusion of overflow capacity. These findings emphasise the importance of transparent reporting in simulation studies and demonstrate how small modelling differences can substantially affect outcomes in highly constrained healthcare systems.

Reference

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Figure 1: CCU patient flowchart

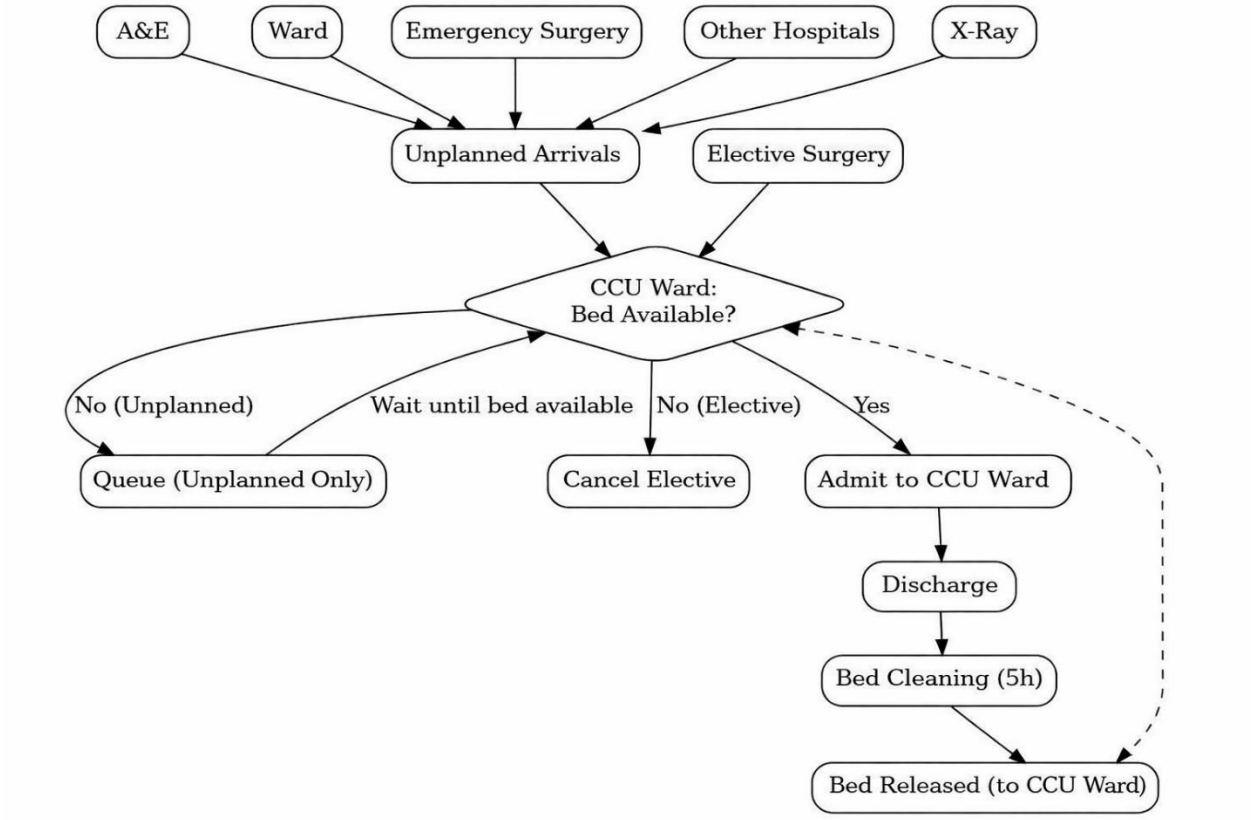

Figure 2: Distribution of CCU Performance metrics across capacity scenarios

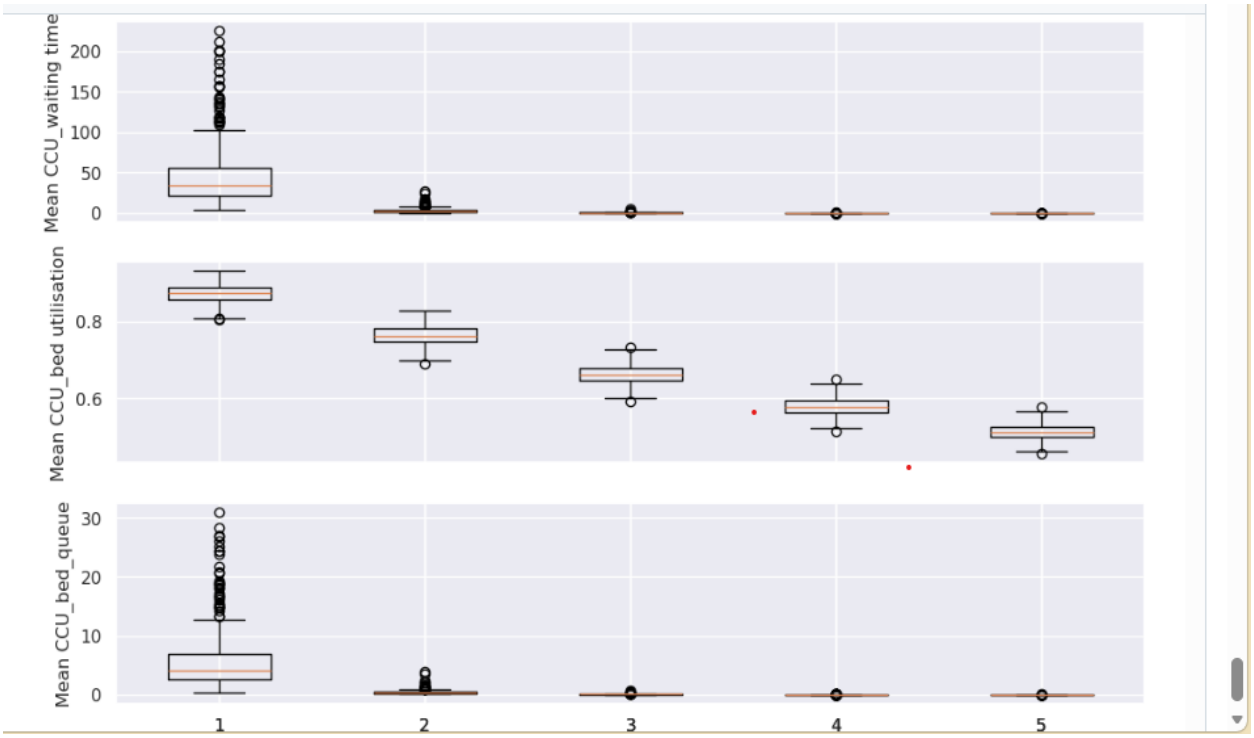

Figure 2: Boxplots of mean waiting time, bed utilisation, and queue size across CCU capacity scenarios. Results show high variability and congestion under baseline capacity, with substantial reductions in delays and queue sizes as bed numbers increase. Utilisation declines progressively with additional capacity, indicating reduced system pressure.